Notes: Kim, Wang & Wang (JFE, 2022)
Geographic clustering of institutional investors

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1 Big picture

- **Question.** Does the geographic concentration of institutional investors matter for local firms, even in a nationally integrated U.S. equity market?
- **Core measure.** The ATM ratio equals the log of one plus local institutional AUM divided by local public-firm market capitalization.
- **Main finding.** Firms headquartered in states with high institutional investor presence have higher market-to-book ratios.
- **Mechanism.** The evidence favors a financing and monitoring channel: local institutions improve access to informed equity capital and reduce financial constraints.
- **Real effects.** High-ATM firms invest more, rely less on cash flow, issue equity more frequently, and local institutions absorb more newly issued equity.
- **Identification support.** Relocation tests do not support sorting toward high-ATM states, and the dot-com crash provides a shock to local institutional capital.
- **Takeaway.** Investor geography can matter for firm valuation and real decisions because local institutional capital is an input into financing and monitoring.

2 Data and measurement

2.1 Institutional investor location and 13F data

The paper collects state-level institutional investor locations from SEC 13F filings and matches locations to Thomson Reuters 13F holdings. Because electronic filings begin in 1993, early-period locations are supplemented with Nelson's Directories from 1980–1999. AUM is domestic equity holdings in U.S. common stocks.

Interpretation: The paper needs physical investor location, not just portfolio holdings, because the hypothesis is that proximity affects information acquisition, monitoring, and access to capital.

2.2 Public firm sample and firm location

The firm sample contains U.S. domestic common stocks listed on NYSE, Amex, and Nasdaq from 1980–2017. Financial firms and utilities are excluded. Firm headquarters identify firm location. CRSP supplies stock prices and shares outstanding; COMPUSTAT supplies accounting variables.

Interpretation: Headquarters location is treated as the place where the firm interacts with investors, employees, and local information networks.

2.3 ATM ratio: institutional-investor presence

$$\text{ATM}_{s,t} = \log\left(1 + \frac{AUM_{s,t}}{MarketCap_{s,t}}\right).$$

A high ATM ratio means institutional capital in state s is large relative to the market capitalization of local public firms. The paper also computes MSA, CSA, and Census-region versions.

Interpretation: The ATM ratio measures geographic dislocation between the demand side of equity capital and the supply side of local public equities.

2.4 Local preference measures

LP1 compares the local-firm weight in local institutions' portfolios to the local-firm weight in the market portfolio. LP2 compares the local-firm weight in local institutions' portfolios to the local-firm weight in all institutions' portfolios.

Interpretation: Local preference and institutional capacity are different. The paper shows that the size of the local institutional base, not local preference intensity, drives valuation.

2.5 Outcomes and controls

The main valuation outcome is market-to-book ratio. Investment outcomes include INVESTMENT, CAPXRND, R&D, and CAPEX. Financing outcomes include equity issuance and debt issuance. Controls include R&D-to-sales, CAPEX-to-sales, ROE, leverage, log assets, state income growth, state population growth, Q, cash flow, and returns.

Interpretation: The controls separate institutional geography from firm size, profitability, investment opportunities, leverage, and local economic growth.

3 Models and empirical specifications

3.1 Baseline valuation regression

$$\mathrm{MB}_{i,s,t} = \alpha + \beta \mathrm{ATM}_{s,t} + \gamma' X_{i,t} + \delta' Z_{s,t} + FE_{industry} + FE_{exchange} + FE_{year} + \varepsilon_{i,s,t}.$$

Interpretation: The coefficient β compares valuation for firms located in states with larger or smaller institutional bases, conditional on firm controls, state controls, industry, exchange, and year effects.

3.2 Local preference and institutional size

$$\mathrm{MB}_{i,s,t} = \alpha + \beta_1 \mathrm{ATM}_{s,t} + \beta_2 \mathrm{LP}_{s,t} + Controls + FE + \varepsilon_{i,s,t}.$$

Interpretation: This separates institutional capacity from local overweighting. A positive ATM coefficient with weak or negative LP effects means the size of the local investor base matters more than the local-bias intensity.

3.3 Equity-dependence subsample tests

$$\mathrm{MB}_{i,s,t} = \alpha + \beta \mathrm{ATM}_{s,t} + Controls + FE + \varepsilon_{i,s,t}$$

separately for small/large firms, high/low KZ firms, payout/no-payout firms, and financially dependent/non-dependent firms.

Interpretation: If local institutional investors alleviate financing frictions, the ATM effect should be stronger for firms more likely to need external equity.

3.4 Investment and financing equations

$$Investment_{i,s,t} = \alpha + \beta \mathrm{ATM}_{s,t} + \theta_1 (Q_{i,t} \times \mathrm{ATM}_{s,t}) + \theta_2 (CashFlow_{i,t} \times \mathrm{ATM}_{s,t}) + Controls + FE + \varepsilon.$$

$$Issuance_{i,s,t} = \alpha + \beta \mathrm{ATM}_{s,t} + \theta_1 (Q_{i,t} \times \mathrm{ATM}_{s,t}) + \theta_2 (CashFlow_{i,t} \times \mathrm{ATM}_{s,t}) + Controls + FE + \varepsilon.$$

Interpretation: The investment equation tests whether high-ATM firms are less cash-flow constrained and more responsive to growth opportunities. The issuance equation tests whether the mechanism is equity access rather than debt access.

3.5 Discount-rate and shock tests

$$Return_{i,t} = \alpha_t + \beta \mathrm{ATM}_{s,t-1} + Controls + \varepsilon_{i,t}.$$

$$\mathrm{MB}_{i,s,t} = \alpha + \beta_1 AffectedState_s + \beta_2 ITBurst_t + \beta_3 (AffectedState_s \times ITBurst_t) + Controls + FE + \varepsilon.$$

Interpretation: Stock-return regressions test the discount-rate channel. The dot-com shock tests whether an adverse shock to local institutions lowers non-tech firm valuations.

4 Identification and empirical design

4.1 Core source of variation

The paper exploits variation in the local institutional investor base relative to local public-firm equity supply. This mismatch is persistent but varies across states and over time.

Interpretation: The design asks whether firms in regions with a deeper local institutional base have higher valuation and better financing outcomes.

4.2 Baseline identifying assumption

The baseline regressions require that, after firm controls, state controls, and fixed effects, the ATM ratio is not merely proxying for unobserved local growth or firm quality. The authors therefore add mechanism and shock tests rather than relying only on cross-sectional panel regressions.

4.3 Mechanism-based identification

If the mechanism is access to informed equity capital, the ATM ratio should predict investment and equity issuance, reduce cash-flow dependence, and lead local institutions to absorb new equity. Tables 5–7 test these implications.

Interpretation: This is a strong internal-consistency check: the same variable that predicts valuation also predicts financing behavior through the channel the paper proposes.

4.4 Discount-rate channel test

If high ATM merely means local investor demand pushes prices up and expected returns down, ATM should predict stock returns. Table 8 finds no robust relation between ATM and returns.

Interpretation: This weakens the discount-rate explanation and supports the financing/monitoring interpretation.

4.5 Relocation tests

Firms may choose headquarters based on institutional investor presence. Tables 9–10 examine firms that move across states and ask whether they move toward high-ATM states or whether ATM predicts relocation patterns.

Interpretation: The results do not support location matching as the main driver of the valuation results.

4.6 Dot-com shock test

The dot-com crash damaged institutions in states heavily exposed to technology stocks. The paper studies non-tech firms in those states before and after the crash. The negative affected-state $\times$ IT-burst coefficient in Table 11 supports a causal link from local institutional capital to firm value.

Interpretation: This is the closest quasi-experimental evidence in the paper: an adverse shock to local institutions reduces nearby non-tech firms' valuation.

4.7 Threats and remaining caveat

The paper cannot perfectly separate institutional presence from local human capital, financial agglomeration, and regional growth. The most persuasive evidence is the combination of valuation regressions, financing-channel tests, relocation tests, return tests, and the dot-com shock.

5 Tables and figures

5.1 Figure 1: The ratio of AUM of 13F institutions in a state to AUM of All U.S. 13F institutions; Figure 2: The ratio of aggregate market capitalization of firms in a state to aggregate market capitalization of All U.S. firms

Read: Figure 1 shows institutional AUM is concentrated in a few states. Figure 2 shows public-firm market capitalization is also concentrated, but the leading states differ from investor clusters.

Economic message: The visual mismatch motivates the paper's central variable: institutional capital and public firms are not located in the same proportions.

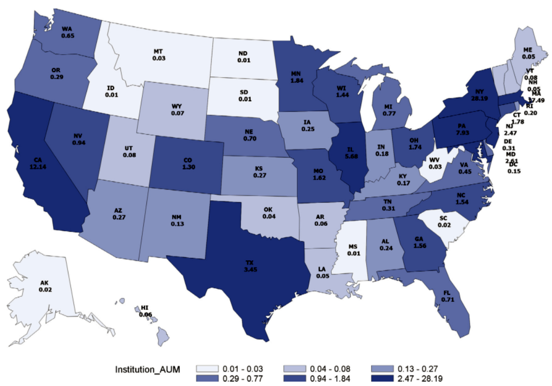

Fig. 1. The ratio of AUM of 13F institutions in a state to AUM of All U.S. 13F institutions. This figure presents a map of the time-series average of the ratio of aggregate AUM of state institutions to the aggregate AUM of all institutions for each state.

(a) Original Figure 1. State share of institutional AUM. Darker states have larger institutional AUM shares.

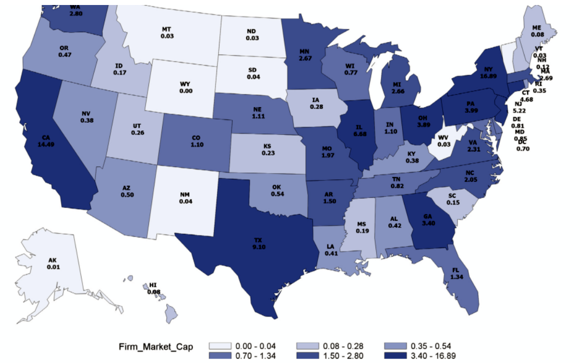

Fig. 2. The ratio of aggregate market capitalization of firms in a state to aggregate market capitalization of All U.S. firms. This figure presents a map of the time-series average of the ratio of the aggregate market capitalization of state firms to the aggregate market capitalization of all firms for each state.

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(b) Original Figure 2. State share of public-firm market capitalization. The map differs from Figure 1, showing geographic dislocation.

5.2 Figure 3: The ratio of aggregate AUM of state institutions to aggregate market capitalization of state firms; Table 1: Summary statistics of state-level variables

Read: Figure 3 maps raw ATM ratios. Table 1 shows institutional AUM grows dramatically from 1980 to 2017 and documents strong cross-state heterogeneity.

Economic message: The key object is local institutional capital relative to local public equity supply, not state size alone.

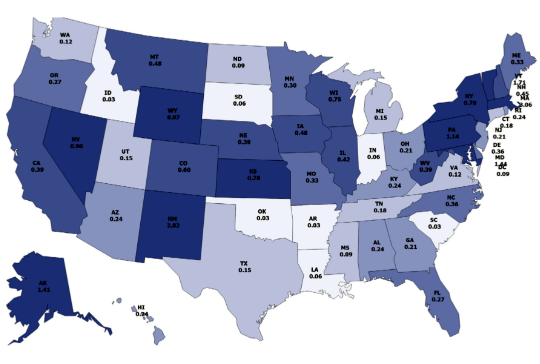

Fig. 3. The ratio of aggregate AUM of state institutions to aggregate market capitalization of state firms. This figure presents a map of the time-series average of the ratio of the aggregate AUM of state institutions to the aggregate market capitalization of state firms for each state.

(a) Original Figure 3. Raw ATM ratio by state.

Table 1
Summary statistics of state-level variables.
This table reports the summary statistics for the state aggregate domestic equity holdings (AUM) of 13F institutions, the number of 13F institutions, the number of public firms, the state aggregate market capitalization of firms, ATM ratio, and two local-preference measures for 1980, 1999, and 2017. The summary statistics are reported for the top 5 and bottom 5 states ranked by the state-level equity holdings. ATM ratio is defined as the logarithm of one plus the ratio of the state aggregate AUM of institutions to the total market capitalization of public firms in the same state. The first local-preference measure (LP1) is defined as the difference between the weight of the local firms in the aggregate portfolio of local institutions and the weight of local firms in the market portfolio. The second local-preference measure (LP2) is defined as the difference between the weight of the local firms in the aggregate portfolio of local institutions and the weight of the local firms in the aggregate portfolio of all 13F institutions.

	State AUM (in $ billion)	Number of institutions	State market cap (in $ billion)	Number of firms	AUM/Market CAP	ATM ratio	LP1	LP2
1980								
National aggregate	209.11	231	1178.40	4294	0.177	0.163		
State average	6.97	8	39.28	84	0.163	0.118	1.240	1.447
State median	1.38	4	13.35	35	0.030	0.030	1.106	1.221
Top 5 states								
NY	68.68	62	240.83	580	0.285	0.251	−0.042	0.163
MA	31.66	29	31.79	158	0.996	0.691	0.709	0.694
CA	19.49	26	147.89	496	0.132	0.124	0.320	0.457
PA	14.97	10	67.15	199	0.223	0.201	1.201	1.279
IL	14.34	16	101.39	226	0.141	0.132	0.643	0.692
Bottom 5 states								
SD	0.00	0	0.12	5	0.000	0.000	.	.
WY	0.00	0	0.47	15	0.000	0.000	.	.
ID	0.00	0	2.58	11	0.000	0.000	.	.
ND	0.00	0	0.25	4	0.000	0.000	.	.
HI	0.00	0	1.50	13	0.000	0.000	.	.
1999								
National aggregate	7382.94	1591	15575.06	6522	0.474	0.388		
State average	144.76	31	305.39	128	0.920	0.419	1.487	1.616
State median	17.72	10	84.97	52	0.252	0.225	1.067	1.262
Top 5 states								
NY	1942.28	382	2380.01	607	0.816	0.597	0.022	0.089
MA	1506.72	128	540.23	354	2.789	1.332	0.160	0.158
CA	997.28	216	3135.41	1108	0.318	0.276	0.053	0.149
PA	535.15	73	388.68	292	1.377	0.866	0.262	0.265
IL	351.38	84	631.74	268	0.556	0.442	0.625	0.684
Bottom 5 states								
AR	0.82	5	347.61	30	0.002	0.002	2.418	2.813
SC	0.42	2	21.03	45	0.020	0.020	3.715	3.787
ID	0.35	2	39.89	16	0.009	0.009	1.439	1.345
SD	0.27	2	25.99	9	0.010	0.010	1.162	1.337
ND	0.23	1	3.15	6	0.073	0.070	1.723	2.038
2017								
National aggregate	17271.38	3190	27213.08	3487	0.635	0.491		
State average	338.65	63	533.59	68	0.897	0.458	0.984	1.161
State median	37.23	20	141.08	27	0.279	0.246	0.636	0.698
Top 5 states								
NY	4947.22	756	3424.90	318	1.444	0.894	−0.029	−0.031
MA	3351.11	230	766.40	222	4.373	1.681	0.299	0.149
PA	2238.91	145	801.22	150	2.794	1.334	0.047	−0.023
CA	1810.14	404	6044.27	611	0.299	0.262	0.023	0.035
IL	969.30	178	1563.71	150	0.620	0.482	0.313	0.302
Bottom 5 states								
ID	1.73	8	65.00	8	0.027	0.026	0.346	0.268
WY	1.67	5	0.33	1	4.989	1.790	0.000	0.000
NV	1.14	3	140.06	30	0.008	0.008	−0.538	−0.342
HI	1.04	4	18.25	12	0.057	0.056	3.804	3.979
ND	0.66	2	6.10	3	0.108	0.103	0.000	0.000

(b) Original Table 1. State-level AUM, firm market cap, ATM, LP1, and LP2.

5.3 Table 2: ATM ratio and firm valuation; Table 3: ATM ratio, local preference, and firm valuation

Read: Table 2 reports the baseline positive relation between ATM and market-to-book. Table 3 shows ATM remains positive after local-preference measures are included.

Economic message: The size of the local institutional base, not merely local-bias intensity, drives valuation.

ATM ratio and firm valuation.
This table reports panel regressions of market-to-book ratio (MB) on ATM ratio, firm-level and state-level control variables, as well as the exchange, year, and industry fixed effects. Industries are defined using 2-digit SIC codes. MB is defined as the ratio of the market value of equity to the book value of equity. ATM ratio is defined as the log of one plus the ratio of the aggregate AUM of institutions in a state to the total market capitalization of public firms in the same state. HKS ratio is defined as the total book value of all firms located in state *i* to the aggregate income of all households located in state *i*. R&D-to-sales is defined as research and development expenditures scaled by sales. R&D is set to zero when it is missing in the data. CAPEX-to-sales is defined as capital expenditures scaled by sales. ROE is defined as net income divided by lagged book equity. Leverage is defined as the sum of long-term debt and debt in current liabilities scaled by lagged assets. Log(Assets) is defined as the log of total assets. PIPCgrowth is defined as the state-level personal income per-capita growth rate. Popgrowth is defined as the state-level population growth rate. Census region (MSA or CSA) ATM ratio is measured at the region (MSA or CSA) level. The sample period is from 1980 to 2017. The *t*-statistics, based on standard errors clustered at the state-year and firm levels, are reported in parentheses. Statistical significance is denoted by *, **, and *** at the 10%, 5%, and 1% levels, respectively.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	(13)
			MSA	CSA	Census region	State subsample				Region subsample			
Variable	State ATM ratio		ATM ratio	ATM ratio	ATM ratio	1980–1998		1999–2017		1980–1998		1999–2017	
ATM ratio	0.448***	0.447***	0.306*	0.413**	0.782***	0.155	0.147	0.529**	0.538**	0.331	0.565**	0.810**	0.824**
	(2.68)	(2.66)	(1.79)	(2.07)	(2.84)	(1.07)	(1.01)	(2.34)	(2.37)	(1.25)	(2.00)	(2.27)	(2.32)
HKS ratio		−0.011	0.213***	0.285**	−0.428		−0.118		0.135		−0.658**		−0.228
		(−0.07)	(2.81)	(2.27)	(−1.39)		(−0.92)		(0.54)		(−2.37)		(−0.56)
R&D-to-sales	0.039*	0.039*	0.016	0.007	0.040	0.677***	0.676***	0.044**	0.044**	0.679***	0.677***	0.045*	0.045*
	(1.70)	(1.70)	(0.77)	(0.29)	(1.56)	(7.03)	(7.02)	(2.00)	(2.00)	(5.84)	(5.83)	(1.84)	(1.84)
CAPEX-to-sales	0.143*	0.143*	0.015	0.062	0.150*	0.074	0.075	−0.312*	−0.313*	0.073	0.076	−0.299	−0.299
	(1.78)	(1.78)	(0.12)	(0.46)	(1.81)	(0.94)	(0.94)	(−1.74)	(−1.75)	(0.90)	(0.93)	(−1.59)	(−1.59)
ROE	−1.529***	−1.529***	−1.504***	−1.609***	−1.512***	−1.516***	−1.516***	−1.479***	−1.479***	−1.505***	−1.504***	−1.456***	−1.456***
	(−10.44)	(−10.44)	(−9.15)	(−8.66)	(−9.54)	(−13.24)	(−13.24)	(−6.24)	(−6.24)	(−13.38)	(−13.37)	(−5.64)	(−5.64)
Leverage	3.097***	3.097***	3.635***	3.695***	3.095***	1.575***	1.576***	5.177***	5.176***	1.585***	1.595***	5.157***	5.157***
	(11.47)	(11.47)	(9.55)	(9.04)	(10.10)	(9.02)	(9.03)	(9.09)	(9.09)	(9.03)	(9.09)	(7.90)	(7.91)
Log(Assets)	−0.266***	−0.266***	−0.230***	−0.229***	−0.266***	−0.402***	−0.402***	−0.149***	−0.149***	−0.403***	−0.404***	−0.148***	−0.149***
	(−8.42)	(−8.42)	(−6.40)	(−6.29)	(−7.91)	(−12.99)	(−12.96)	(−2.93)	(−2.93)	(−11.68)	(−11.70)	(−2.88)	(−2.89)
PIPCgrowth	5.920***	5.948***	9.793***	9.053***	7.016**	2.100	2.416	12.372***	12.062***	−1.451	−0.552	16.593***	17.338***
	(2.92)	(2.88)	(4.61)	(3.09)	(2.43)	(1.12)	(1.30)	(3.67)	(3.54)	(−0.67)	(−0.27)	(3.40)	(3.36)
Popgrowth	3.489	3.383	−0.634	1.627	1.532	8.036**	7.001**	−7.646	−6.152	7.100*	−0.281	2.520	−1.409
	(0.81)	(0.76)	(−0.14)	(0.30)	(0.25)	(2.35)	(2.08)	(−0.67)	(−0.54)	(1.73)	(−0.06)	(0.17)	(−0.09)
N	109,490	109,490	88,265	87,610	109,256	56,715	56,715	52,772	52,772	56,611	56,611	52,642	52,642
Adjusted R^2	0.069	0.069	0.065	0.063	0.069	0.110	0.110	0.065	0.065	0.110	0.111	0.065	0.065
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Exchange FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Industry FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Table 3
ATM ratio, local preference, and firm valuation.
This table reports panel regressions of market-to-book ratio (MB) on ATM ratio, local preference, firm-level and state-level control variables, as well as the exchange, year, and industry fixed effects. Industries are defined using 2-digit SIC codes. MB is defined as the ratio of the market value of equity to the book value of equity. ATM ratio is defined as the log of one plus the ratio of the aggregate AUM of institutions in a state to the total market capitalization of public firms in the same state. LP1 is defined as the log of the ratio of the aggregate weight of local firms in the portfolio of local institutions to the aggregate weight of local firms in the market portfolio. LP2 is defined as the log of the ratio of the aggregate weight of local firms in the portfolio of local institutions to the aggregate weight of local firms in the portfolio of all institutions. R&D-to-sales is defined as research and development expenditures scaled by sales. R&D is set to zero when it is missing in the data. CAPEX-to-sales is defined as capital expenditures scaled by sales. ROE is defined as net income divided by lagged book equity. Leverage is defined as the sum of long-term debt and debt in current liabilities scaled by lagged assets. Log(Assets) is defined as the log of total assets. PIPCgrowth is defined as the state-level personal income per-capita growth rate. Popgrowth is defined as the state-level population growth rate. Census region ATM ratio is measured at the region level. The sample period is from 1980 to 2017. The *t*-statistics, based on standard errors clustered at the state-year and firm levels, are reported in parentheses. Statistical significance is denoted by *, **, and *** at the 10%, 5%, and 1% levels, respectively.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Variable	State				Region			
LP1	−0.132***		−0.103***		−0.256**		−0.158	
	(−3.51)		(−2.61)		(−2.26)		(−1.31)	
LP2		−0.132***		−0.099**		−0.276**		−0.147
		(−3.55)		(−2.56)		(−2.36)		(−1.16)
ATM ratio			0.382**	0.377**			0.646**	0.636**
			(2.21)	(2.17)			(2.26)	(2.18)
R&D-to-sales	0.039*	0.039*	0.037	0.037	0.041	0.041	0.040	0.040
	(1.71)	(1.71)	(1.64)	(1.64)	(1.60)	(1.59)	(1.57)	(1.56)
CAPEX-to-sales	0.159*	0.160*	0.157*	0.158*	0.145*	0.148*	0.145*	0.147*
	(1.84)	(1.87)	(1.82)	(1.84)	(1.76)	(1.81)	(1.75)	(1.79)
ROE	−1.526***	−1.526***	−1.524***	−1.524***	−1.512***	−1.511***	−1.511***	−1.511***
	(−10.33)	(−10.33)	(−10.33)	(−10.33)	(−9.54)	(−9.53)	(−9.53)	(−9.53)
Leverage	3.075***	3.075***	3.090***	3.090***	3.089***	3.090***	3.095***	3.095***
	(11.25)	(11.25)	(11.26)	(11.26)	(10.07)	(10.07)	(10.09)	(10.09)
Log(Assets)	−0.265***	−0.265***	−0.265***	−0.265***	−0.265***	−0.266***	−0.264***	−0.264***
	(−8.32)	(−8.32)	(−8.30)	(−8.31)	(−8.02)	(−8.04)	(−7.84)	(−7.85)
PIPCgrowth	6.938***	6.783***	6.465***	6.350***	6.252**	5.888**	5.950**	5.762**
	(3.39)	(3.32)	(3.11)	(3.06)	(2.29)	(2.17)	(2.19)	(2.14)
Popgrowth	−2.565	−2.284	1.104	1.325	0.616	1.638	5.667	6.237
	(−0.55)	(−0.49)	(0.24)	(0.29)	(0.11)	(0.30)	(1.09)	(1.21)
N	107,988	107,988	107,988	107,988	109,256	109,256	109,256	109,256
Adjusted R^2	0.068	0.068	0.068	0.068	0.068	0.068	0.068	0.068
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Exchange FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Industry FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes

as financially constrained (Hadlock and Pierce, 2010). A firm is classified as small if its market capitalization is below the 50% NYSE size breakpoints. Second, we rank firms based on their KZ index in the previous year. A firm is classified as financially constrained if its KZ index is above the 50% KZ index breakpoints.[14] Third, we consider firms with lower payout ratios (Fazzari et al., 1988) financially constrained. A firm is classified as financially constrained if its payout ratio is 0. We define the payout ratio as the ratio of total distributions (dividends plus purchase of com- payout, no S&P crediting rating or S&P rating below BBB-, and the ratio of net sales to lagged assets not in the top quintile in the previous year among sample firms.

The results for the subsamples are displayed in Table 4. For the size subsamples, the coefficient of the ATM ratio for small firms is 0.384, which is significant at the 1% level. The coefficient of ATM ratio for large firms is 0.610, but it is insignificant. Columns (3) and (4) are the results based on the KZ index. The influence of the ATM ratio on valuation is significantly positive (0.646) for high KZ index

5.4 Table 4: ATM ratio and firm valuation: subsample analysis

Read: The ATM effect is stronger for equity-dependent firms, including small, high-KZ, low-payout, and financially dependent firms.

Economic message: Local institutional capital is most valuable where external equity access is most important.

5.5 Table 5: ATM ratio and corporate investment; Table 6: ATM ratio and corporate financing decisions; Table 7: Local institutional holding of newly issued equity

Read: Table 5 shows high-ATM firms invest more and are less cash-flow dependent. Table 6 shows high-ATM firms issue more equity but not more debt. Table 7 shows local institutions hold more newly issued local equity.

Economic message: These tables provide the financing-channel mechanism: local institutions help firms finance investment with equity.

Table 4
ATM ratio and firm valuation: subsample analysis.
This table reports panel regressions of market-to-book ratio (MB) on ATM ratio, firm-level and state-level control variables, as well as the exchange, year, and industry fixed effects. Industries are defined using 2-digit SIC codes. The subsamples of firms are defined using size, KZ Index, payout ratio, and a financial-dependence measure. MB is defined as the ratio of the market value of equity to the book value of equity. ATM ratio is defined as the log of one plus the ratio of the aggregate AUM of institutions in a state to the total market capitalization of public firms in the same state. R&D-to-sales is defined as research and development expenditures scaled by sales. R&D is set to zero when it is missing in the data. CAPEX-to-sales is defined as capital expenditures scaled by sales. ROE is defined as net income divided by lagged book equity. Leverage is defined as the sum of long-term debt and debt in current liabilities scaled by lagged assets. Log(Assets) is defined as the log of total assets. PIPCgrowth is defined as the state-level personal income per-capita growth rate. Popgrowth is defined as the state-level population growth rate. The sample period is from 1980 to 2017. The t-statistics, based on standard errors clustered at the state-year and firm levels, are reported in parentheses. Statistical significance is denoted by *, **, and *** at the 10%, 5%, and 1% levels, respectively.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	Size		KZ index		Payout ratio		Financial dependence	
Variable	Small	Large	Top 50%	Bottom 50%	No Payout	Payout	Dependent	Independent
ATM ratio	0.384***	0.610	0.646**	0.288*	0.400**	0.538*	0.387*	0.489**
	(2.60)	(1.22)	(2.37)	(1.65)	(2.36)	(1.83)	(1.70)	(2.21)
R&D-to-sales	0.039*	0.077*	0.161***	0.022	0.017	0.342***	−0.014	0.205***
	(1.80)	(1.95)	(3.45)	(1.09)	(0.74)	(2.84)	(−0.61)	(2.68)
CAPEX-to-sales	0.174**	−2.470***	−0.151	0.313***	0.146*	−0.133	0.142	−0.058
	(2.20)	(−4.81)	(−1.16)	(3.49)	(1.81)	(−0.53)	(1.54)	(−0.39)
ROE	−1.698***	−0.834	−2.008***	−0.910***	−1.675***	−1.217***	−1.921***	−1.141***
	(−13.01)	(−1.46)	(−10.05)	(−5.90)	(−11.06)	(−3.64)	(−9.96)	(−6.07)
Leverage	2.868***	6.382***	4.404***	3.405***	2.905***	3.742***	2.941***	3.247***
	(13.24)	(7.32)	(11.74)	(6.83)	(11.37)	(7.21)	(7.75)	(10.16)
Log(Assets)	−0.662***	−0.929***	−0.426***	−0.144***	−0.400***	−0.144***	−0.504***	−0.172***
	(−19.85)	(−7.87)	(−10.08)	(−4.05)	(−11.52)	(−2.99)	(−10.36)	(−4.75)
PIPCgrowth	6.841***	3.171	3.830	7.469***	8.513***	3.323	11.823***	3.206
	(3.12)	(0.75)	(1.32)	(3.76)	(3.10)	(1.35)	(3.43)	(1.55)
Popgrowth	7.776**	−17.610	0.231	5.944	9.012*	−2.287	9.066	1.136
	(2.00)	(−1.42)	(0.04)	(1.29)	(1.85)	(−0.36)	(1.45)	(0.22)
N	84,926	24,152	54,661	54,563	59,975	49,512	33,279	76,210
Adjusted R^2	0.087	0.102	0.082	0.073	0.078	0.067	0.095	0.060
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Exchange FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Industry FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes

ically, we examine the link between the ATM ratio and corporate investment and financing decisions.

The high firm valuation in states with greater institutional presence could be driven by the lowered discount rate due to excess local institutional investor demand for local equity, similar to the individual investor "only-game-in-town" effect proposed in Hong et al. (2008). However, the high valuation could be supported by the lowered financing friction because of a more informed investor base and reduced information asymmetry. Studies that document institutional investor local preference also find these investors benefit from local information advantage (Coval and Moskowitz, 1999; Baik et al., 2010). Local institutional investors are also effective monitors of corporate behavior increasing institutional ownership increases management disclosure and helps reduce information asymmetry. Several recent studies highlight the role of local institutional investors in engaging in monitoring activities and improving corporate governance (see, e.g., Kang and Kim, 2008; Gaspar and Massa, 2007). Different from these papers that study the heterogeneity of institutional ownership across individual firms, we examine whether the state-level institutional investor presence has a real effect on the policies of local companies.

4.1. ATM ratio and corporate investment

We first examine the link between institutional investor

Table 5
ATM ratio and corporate investment.
This table reports panel regressions of investment on ATM ratio, two interactions terms, firm-level and state-level control variables, as well as the exchange, year, and industry fixed effects. Industries are defined using 2-digit SIC codes. INVESTMENT is the sum of total asset growth and R&D expenditures. CAPXRND is the sum of capital expenditures and R&D expenditures. R&D is research and development expenditures. CAPEX is capital expenditures. The four investment variables are scaled by lagged total assets. ATM ratio is defined as the log of one plus the ratio of the aggregate AUM of institutions in a state to the total market capitalization of public firms in the same state. Cash flow is defined as net income before extraordinary items, depreciation and amortization expenses, scaled by lagged assets. Q is defined as market value of equity plus total assets, less book value of equity, scaled by total assets. StateQ is defined as value-weighted Q of firms within each state. ROE is defined as net income divided by lagged book equity. Log(Assets) is defined as the log of total assets. Leverage is defined as the sum of long-term debt and debt in current liabilities scaled by lagged assets. Ret is the stock return in the previous year. PIPCgrowth is defined as the state-level personal income per-capita growth rate. Popgrowth is defined as the state-level population growth rate. The sample period is from 1980 to 2017. The t-statistics, based on standard errors clustered at the state-year and firm levels, are reported in parentheses. Statistical significance is denoted by *, **, and *** at the 10%, 5%, and 1% levels, respectively.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
Variable		INVESTMENT			CAPXRND			R&D			CAPEX	
ATM ratio	0.016***	−0.003	0.014**	0.014***	0.005	0.013***	0.016***	−0.002	0.015***	−0.002	0.006***	−0.002
	(2.72)	(−0.25)	(2.38)	(4.35)	(1.18)	(4.16)	(5.46)	(−0.58)	(5.43)	(−1.19)	(2.74)	(−1.36)
Q×ATM ratio		0.009			0.004*			0.008***			−0.003***	
		(1.36)			(1.83)			(4.28)			(−4.34)	
Cash flow×ATM ratio			−0.252***			−0.152***			−0.113***			−0.040***
			(−5.68)			(−10.15)			(−8.80)			(−6.89)
Cash flow	0.051**	0.052**	0.157***	−0.137***	−0.137***	−0.074***	−0.181***	−0.180***	−0.133***	0.054***	0.054***	0.071***
	(2.05)	(2.09)	(4.56)	(−16.21)	(−16.13)	(−6.19)	(−28.69)	(−28.38)	(−14.93)	(15.15)	(15.11)	(14.50)
Q	0.090***	0.087***	0.090***	0.024***	0.022***	0.024***	0.014***	0.011***	0.014***	0.008***	0.009***	0.008***
	(36.27)	(25.54)	(36.20)	(21.81)	(15.88)	(21.98)	(21.40)	(12.29)	(21.59)	(16.43)	(13.72)	(16.48)
StateQ	0.002	0.002	0.002	0.002***	0.002***	0.002***	0.004***	0.004***	0.004***	−0.001***	−0.001***	−0.001***
	(1.23)	(1.07)	(1.18)	(2.96)	(2.79)	(2.83)	(4.50)	(3.97)	(4.32)	(−4.02)	(−3.44)	(−4.13)
ROE	−0.007	−0.007	−0.007	−0.009***	−0.009***	−0.009***	−0.009***	−0.009***	−0.009***	0.001	0.001	0.001
	(−1.55)	(−1.57)	(−1.53)	(−7.13)	(−7.15)	(−7.13)	(−9.05)	(−9.12)	(−9.01)	(0.73)	(0.76)	(0.75)
Log(Assets)	−0.015***	−0.015***	−0.015***	0.001	0.001	0.000	0.001***	0.001***	0.001***	−0.000	−0.000	−0.000
	(−11.10)	(−11.13)	(−11.37)	(0.92)	(0.88)	(0.61)	(3.26)	(3.12)	(2.95)	(−1.19)	(−1.12)	(−1.34)
Leverage	−0.010	−0.010	−0.009	−0.031***	−0.031***	−0.030***	−0.047***	−0.047***	−0.047***	0.015***	0.015***	0.015***
	(−1.37)	(−1.39)	(−1.19)	(−9.62)	(−9.64)	(−9.31)	(−22.81)	(−23.00)	(−22.30)	(7.33)	(7.35)	(7.41)
Ret	0.112***	0.112***	0.111***	0.015***	0.015***	0.014***	0.003***	0.003***	0.002**	0.011***	0.011***	0.011***
	(23.91)	(23.89)	(23.82)	(11.70)	(11.67)	(11.39)	(2.74)	(2.64)	(2.41)	(15.26)	(15.32)	(15.24)
PIPCgrowth	0.540***	0.543***	0.528***	0.242***	0.244***	0.235***	0.064	0.067	0.059	0.155***	0.154***	0.153***
	(4.16)	(4.18)	(4.11)	(3.63)	(3.65)	(3.54)	(1.24)	(1.29)	(1.12)	(4.59)	(4.54)	(4.60)
Popgrowth	1.011***	1.023***	1.026***	0.949***	0.954***	0.958***	0.249***	0.260***	0.255***	0.674***	0.669***	0.676***
	(4.54)	(4.61)	(4.62)	(7.60)	(7.64)	(7.73)	(2.72)	(2.84)	(2.79)	(9.82)	(9.74)	(9.91)
N	113,005	113,005	113,005	111,877	111,877	111,877	113,013	113,013	113,013	111,877	111,877	111,877
Adjusted R^2	0.199	0.199	0.201	0.316	0.316	0.324	0.503	0.506	0.511	0.233	0.233	0.234
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Exchange FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Industry FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Table 6
ATM ratio and corporate financing decisions.
This table reports panel regressions of equity and debt issuance on ATM ratio, two interaction terms, firm-level and state-level control variables, as well as the exchange, year, and industry fixed effects. Industries are defined using 2-digit SIC codes. Equity issuance is defined as the change in book equity and the change in deferred taxes, less the change in retained earnings, scaled by lagged assets. Debt issuance is defined as the change in assets, less the change in book equity, less the change in deferred taxes, scaled by lagged assets. ATM ratio is defined as the log of one plus the ratio of the aggregate AUM of institutions in a state to the total market capitalization of public firms in the same state. Cash flow is defined as net income before extraordinary items, depreciation and amortization expenses, all scaled by lagged assets. Q is defined as market value of equity plus total assets, less book value of equity, all scaled by total assets. R&D-to-sales is defined as research and development expenditures scaled by sales. ROE is defined as net income divided by lagged book equity. PIPCgrowth is defined as the state-level personal income per-capita growth rate. Popgrowth is defined as the state-level population growth rate. The sample period is from 1980 to 2017. The t-statistics, based on standard errors clustered at the state-year and firm levels, are reported in parentheses. Statistical significance is denoted by *, **, and *** at the 10%, 5%, and 1% levels, respectively.

	(1)	(2)	(3)	(4)	(5)	(6)
Variable		Equity issuance			Debt issuance	
ATM ratio	0.013**	−0.053**	0.009**	−0.003	−0.016*	−0.004
	(2.46)	(−2.51)	(2.01)	(−1.21)	(−1.65)	(−1.41)
Q×ATM ratio		0.029***			0.006	
		(3.03)			(1.29)	
Cash flow×ATM ratio			−0.158***			−0.024
			(−4.85)			(−1.52)
Cash flow	−0.089	−0.088	−0.028	−0.011	−0.010	−0.001
	(−1.35)	(−1.35)	(−0.40)	(−0.86)	(−0.86)	(−0.09)
Q	0.042***	0.033***	0.042***	0.011***	0.010***	0.011***
	(5.62)	(3.76)	(5.72)	(5.87)	(3.33)	(5.93)
R&D-to-sales	0.000	0.000	0.000	0.000	0.000	0.000
	(1.10)	(1.10)	(1.08)	(0.79)	(0.75)	(0.65)
ROE	−0.000***	−0.000***	−0.000***	0.000	0.000	0.000
	(−2.97)	(−3.02)	(−2.87)	(0.27)	(0.26)	(0.27)
PIPCgrowth	0.321***	0.316***	0.315***	0.219***	0.218***	0.218***
	(3.76)	(3.73)	(3.68)	(3.01)	(2.98)	(3.00)
Popgrowth	0.855***	0.881***	0.862***	0.495***	0.500***	0.496***
	(4.71)	(4.87)	(4.77)	(4.62)	(4.64)	(4.62)
N	106,883	106,883	106,883	107,090	107,090	107,090
Adjusted R^2	0.222	0.227	0.226	0.036	0.036	0.036
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Exchange FE	Yes	Yes	Yes	Yes	Yes	Yes
Industry FE	Yes	Yes	Yes	Yes	Yes	Yes

al institutions, the implication is that lo- ences across the comparisons. We divide
ıvest more in newly issued shares than subsamples based on the ATM ratios of t

(a) Original Table 6. Equity and debt issuance.

Table 7
Local institutional holding of newly issued equity.
This table compares holdings of newly issued equity by local institutions against three benchmarks for a subsample of firms that have equity issuance of higher than 3%. Local share of the new issue is defined as the ratio of the newly issued shares held by local institutions to the total number of newly issued shares. This variable is measured against (1) the ratio of the total AUM of local institutions to the total AUM of all institutions, (2) the ratio of the total market capitalization of local firms held by local institutions to the total market capitalization of all local firms, (3) the ratio of the number of pre-issuance shares held by local institutions to the pre-issuance shares outstanding of the firm. The table reports the average difference for all firms and separately for large and small firms for the full sample period. The average is computed for all states as well as for high ATM ratio and low ATM ratio states.

		All firms				Large firms				Small firms			
Local share of the new issue versus	All firms	High ATM state	Low ATM state	Diff. (H-L)	t-stat	High ATM state	Low ATM state	Diff. (H-L)	t-stat	High ATM state	Low ATM state	Diff. (H-L)	t-stat
Local inst. AUM of all institution	0.103	0.113	0.081	0.032	8.03	0.205	0.140	0.064	6.00	0.094	0.067	0.028	6.44
Local ownership of all local firms	0.124	0.144	0.080	0.063	15.66	0.234	0.140	0.094	8.72	0.125	0.066	0.059	13.71
Local ownership of the issuing firms	0.135	0.159	0.083	0.076	18.74	0.220	0.139	0.081	7.34	0.147	0.070	0.077	17.86

argument that the local equity financing channel can at partly driven by local information advantage, individual in-

(b) Original Table 7. Local institutional holdings of new equity.

5.6 Table 8: ATM ratio and stock returns

Read: The ATM coefficient is not significant in Fama–MacBeth return regressions.

Economic message: The valuation effect is unlikely to be a pure discount-rate effect.

Table 8
ATM ratio and stock returns.
This table reports results of the monthly Fama–Macbeth regressions of stock returns on ATM ratio, firm-level, and state-level control variables. ATM ratio is defined as the log of one plus the ratio of the aggregate AUM of institutions in a state to the total market capitalization of public firms in the same state. HKS ratio is defined as the total book value of all firms located in state *i* to the aggregate income of all households located in state *i*. Beta is estimated by running regression of stock returns on market returns for the previous 36 months. Size is the log of market capitalization. BM is defined as the ratio of book value of equity to market value of equity. Illiquidity is defined as the average of the absolute value of stock return divided by dollar trading volume on a given day within month. Momentum is defined as the cumulative stock return over the previous one year period. PIPCgrowth is defined as the state-level personal income per-capita growth rate. Popgrowth is defined as the state-level population growth rate. The sample period is from 1980 to 2017. Statistical significance is denoted by *, **, and *** at the 10%, 5%, and 1% levels, respectively.

	(1)	(2)	(3)	(4)	(5)	(6)
		State			Region	
ATM ratio	0.000		0.000	−0.000		0.001
	(0.72)		(0.59)	(−0.20)		(0.43)
HKS ratio		−0.001*	−0.001		−0.002	−0.003
		(−1.71)	(−1.60)		(−1.42)	(−1.44)
Beta	0.000	0.000	0.000	0.000	0.000	0.000
	(0.25)	(0.25)	(0.25)	(0.26)	(0.27)	(0.26)
Size	−0.000	−0.000	−0.000	−0.000	−0.000	−0.000
	(−0.96)	(−0.94)	(−0.93)	(−0.95)	(−0.96)	(−0.94)
BM	0.001	0.001	0.001	0.001	0.001	0.001
	(0.89)	(0.87)	(0.89)	(0.85)	(0.84)	(0.85)
Illiquidity	0.172***	0.172***	0.173***	0.173***	0.173***	0.173***
	(3.46)	(3.48)	(3.50)	(3.48)	(3.48)	(3.49)
Momentum	0.002	0.002	0.002	0.002	0.002	0.002
	(1.07)	(1.03)	(1.04)	(1.03)	(1.02)	(1.01)
PIPCgrowth	0.076***	0.093***	0.083***	0.094*	0.110**	0.096*
	(2.95)	(3.25)	(3.00)	(1.88)	(2.32)	(1.81)
Popgrowth	−0.011	−0.032	−0.022	−0.004	−0.025	−0.018
	(−0.30)	(−0.91)	(−0.58)	(−0.09)	(−0.43)	(−0.32)
N	420	420	420	420	420	420
Average R^2	0.037	0.037	0.037	0.037	0.037	0.038

to capital and their more efficient   *5.1. Firm relocation*

Original Table 8. ATM ratio and stock returns.

5.7 Table 9: ATM ratio and firm relocation; Table 10: Firm relocation and changes in local institutional holdings

Read: Table 9 does not support firms relocating toward high-ATM states. Table 10 examines how local institutional ownership changes around relocation.

Economic message: The relocation evidence reduces concern that high-valuation firms self-select into high-ATM locations.

Table 9
ATM ratio and firm relocation.
This table reports panel regressions of ATM ratio change or ATM ratio change dummy on the KZ index (KZ dummy), firm-level and state-level control variables, as well as the exchange, year, and industry fixed effects. Industries are defined using 2-digit SIC codes. ATM ratio change is defined as the difference of ATM ratio of a firm's new location 1 year after relocation and ATM ratio of a firm's old location 1 year before relocation. Positive ATM ratio change dummy is set to one if ATM ratio change is positive, zero otherwise. KZ dummy is set to one if a firm's KZ index is higher than the top 20% breakpoints, zero otherwise. ATM ratio is defined as the log of one plus the ratio of the aggregate AUM of institutions in a state to the total market capitalization of public firms in the same state. Cash flow is defined as net income before extraordinary items, depreciation and amortization expenses, scaled by lagged assets. *Q* is defined as market value of equity plus total assets, less book value of equity, scaled by total assets. R&D-to-sales is defined as research and development expenditures scaled by sales. ROE is defined as net income divided by lagged book equity. Tax change is defined as the difference of the corporate tax rates between the new location and old location. PIPCgrowth is defined as the state-level personal income per-capita growth rate. Popgrowth is defined as the state-level population growth rate. The sample period is from 1980 to 2017. The *t*-statistics, based on standard errors clustered at the state-year and firm levels, are reported in parentheses. Statistical significance is denoted by *, **, and *** at the 10%, 5%, and 1% levels, respectively.

	(1)	(2)	(3)	(4)
	ΔATM ratio		Positive ATM ratio change dummy	
KZ index	−0.009	0.006		
	(−0.74)	(0.40)		
KZ dummy			−0.036	−0.090
			(−0.62)	(−1.55)
Cash flow		0.043***		0.037***
		(3.01)		(2.80)
Q		−0.012		−0.006
		(−1.38)		(−0.79)
R&D-to-sales		0.001		0.000
		(0.39)		(0.12)
ROE		0.004		−0.001
		(1.06)		(−0.16)
Tax change		0.019***		0.032***
		(3.56)		(5.40)
PIPCgrowth		2.597		0.474
		(1.14)		(0.22)
Popgrowth		−10.287***		−5.545
		(−2.79)		(−1.60)
N	501	441	501	441
Adjusted R^2	0.008	0.050	0.013	0.132
Year FE	Yes	Yes	Yes	Yes
Exchange FE	Yes	Yes	Yes	Yes
Industry FE	Yes	Yes	Yes	Yes

itutions pending on the investment policy a

(a) Original Table 9. Relocation regressions.

ıble 10
rm relocation and changes in local institutional holdings.
ıis table reports changes of local institutional holdings of firms that relocate across states. Local institutional holding is defined as the ratio of the local institutional holding to the total institutional holding of firm. We compare the local institutional holding of the firm 1 year before with local holding 1 year to 3 years after firm relocation for both old and new locations. The table reports the average change and ıe portions of increase (decrease) in local institutional holding after firm relocation. The results are reported for the full sample of firm relocations (all moves), the positive ATM move sample (firms move to gher ATM ratio states), and the negative ATM move sample (firms move to lower ATM ratio states).

Panel A: Change of local institution holding at old locations

	All moves						Moves to higher ATM ratio states						Moves to lower ATM ratio states					
	Before	After	Diff	*t*-stat diff	% of Pos. change	% of Neg. change	Before	After	Diff	*t*-stat diff	% of Pos. change	% of Neg. change	Before	After	Diff	*t*-stat diff	% of Pos. change	% of Neg. change
(−1,+1)	0.078	0.073	−0.005	−0.81	46.20%	53.80%	0.069	0.051	−0.017	−1.93	44.62%	55.38%	0.085	0.089	0.004	0.44	47.17%	52.83%
(−1,+2)	0.078	0.071	−0.008	−1.10	47.56%	52.44%	0.069	0.058	−0.011	−0.95	46.72%	53.28%	0.085	0.079	−0.006	−0.63	48.11%	51.89%
(−1,+3)	0.079	0.054	−0.025	−3.69	39.27%	60.73%	0.069	0.037	−0.033	−3.44	37.41%	62.59%	0.086	0.066	−0.020	−2.07	40.47%	59.53%

Panel B: Change of local institution holding at new locations

	All moves						Moves to higher ATM ratio states						Moves to lower ATM ratio states					
	Before	After	Diff	*t*-stat diff	% of Pos. change	% of Neg. change	Before	After	Diff	*t*-stat diff	% of Pos. change	% of Neg. change	Before	After	Diff	*t*-stat diff	% of Pos. change	% of Neg. change
(−1,+1)	0.038	0.060	0.022	4.02	58.28%	41.72%	0.068	0.108	0.039	3.39	57.24%	42.76%	0.017	0.027	0.010	2.21	59.42%	40.58%
(−1,+2)	0.039	0.056	0.018	3.34	55.84%	44.16%	0.068	0.095	0.027	2.49	53.75%	46.25%	0.018	0.029	0.012	2.29	58.11%	41.89%
(−1,+3)	0.038	0.054	0.015	2.55	54.31%	45.69%	0.067	0.080	0.013	1.20	53.85%	46.15%	0.018	0.035	0.017	2.52	54.78%	45.22%

(b) Original Table 10. Institutional holdings around relocation.

5.8 Table 11: Shocks to local institutions and firm valuation; Appendix Table A.1: Variable definitions

Read: Table 11 uses the dot-com crash as a shock to local institutions with high tech exposure. The affected-state by IT-burst interaction is negative for non-tech firms.

Economic message: When local institutions are damaged, nearby non-tech firms lose valuation, supporting a causal role of local institutional capital.

6 Short summary

- The paper studies whether institutional-investor geography affects public firms.
- The ATM ratio measures local institutional AUM relative to local firm market capitalization.
- Firms in high-ATM states have higher valuations.
- The effect is stronger for equity-dependent firms.
- High-ATM firms invest more, rely less on internal cash flow, and issue more equity.
- Local institutions absorb more newly issued local equity.
- ATM does not predict stock returns, weakening the discount-rate channel.
- Relocation tests do not support endogenous matching as the main driver.

Table 11
Shocks to local institutions and firm valuation.
This table reports panel regressions of firm valuation and state institution tech stock exposure during the tech bust of 2000–2002. The dependent variable (MB) is defined as the ratio of the market value of equity to the book value of equity. Affected state is defined as one if the proportion of tech stocks held by state financial institutions in the first quarter of 2000 is ranked at the top 10% among all states, zero otherwise. We compare MB of the top 10% states with that of all the other states, and the bottom 10%, 20%, and 30% states. IT burst is set to one if the time period is between 2000 to 2002. R&D-to-sales is defined as research and development expenditures scaled by sales. CAPEX-to-sales is defined as capital expenditures scaled by sales. ROE is defined as net income divided by lagged book equity. Log(Assets) is defined as the log of total assets. PIPCgrowth is defined as the state-level personal income per-capita growth rate. Popgrowth is defined as the state-level population growth rate. We include exchange, year, and industry fixed effects. Industries are defined using 2-digit SIC codes. The sample period is from 1990 to 2008. The sample excludes the tech industry, industries highly correlated with the tech industry, and firms located in California. The *t*-statistics, based on standard errors clustered at the state-year level, are reported in parentheses. Statistical significance is denoted by *, **, and *** at the 10%, 5%, and 1% levels, respectively.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	Top 10% states against							
Variable	The rest states		Bottom 10% states		Bottom 20% states		Bottom 30% states	
Affected states	0.177	0.177	0.481	0.480	0.487	0.492	0.400	0.407
	(0.79)	(0.80)	(0.78)	(0.77)	(0.92)	(0.93)	(1.27)	(1.29)
Affected states × IT burst	−0.646**	−0.648**	−1.726***	−1.762***	−1.223***	−1.224***	−0.892**	−0.961***
	(−2.33)	(−2.32)	(−3.12)	(−3.07)	(−3.02)	(−3.02)	(−2.54)	(−2.69)
PIPCgrowth	4.905**	5.031**	14.106	13.537	12.197	15.009*	15.547***	19.608***
	(2.36)	(2.13)	(0.95)	(0.85)	(1.62)	(1.76)	(3.91)	(4.28)
PIPCgrowth × IT burst		−0.893		7.019		−19.242*		−24.736***
		(−0.22)		(0.28)		(−1.81)		(−3.45)
R&D-to-sales	8.668***	8.668***	4.319***	4.319***	4.883***	4.906***	4.110***	4.125***
	(8.92)	(8.92)	(3.75)	(3.75)	(2.97)	(2.99)	(4.97)	(4.99)
CAPEX-to-sales	−0.258	−0.258	−0.198	−0.198	−0.051	−0.052	0.136	0.132
	(−1.39)	(−1.39)	(−0.22)	(−0.22)	(−0.16)	(−0.16)	(0.32)	(0.31)
ROE	1.623***	1.623***	2.425***	2.425***	2.039***	2.032***	2.057***	2.057***
	(9.29)	(9.29)	(2.87)	(2.87)	(3.98)	(3.96)	(5.53)	(5.53)
Leverage	−0.170***	−0.170***	−0.391**	−0.391**	−0.339***	−0.340***	−0.247***	−0.249***
	(−8.50)	(−8.50)	(−2.17)	(−2.17)	(−4.17)	(−4.19)	(−4.94)	(−4.98)
Log(Assets)	−0.032	−0.032	−0.005	−0.005	0.011	0.009	−0.117	−0.119
	(−0.25)	(−0.25)	(−0.16)	(−0.16)	(0.14)	(0.12)	(−0.74)	(−0.75)
Popgrowth	−1.651	−1.670	12.108	12.291	3.955	3.267	2.110	0.913
	(−0.60)	(−0.60)	(0.90)	(0.92)	(0.45)	(0.37)	(0.41)	(0.18)
N	27,640	27,640	2310	2310	4210	4210	7756	7756
Adjusted R^2	0.063	0.063	0.065	0.065	0.089	0.089	0.107	0.107
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Exchange FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Industry FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Table A.1
Variable definitions.

This table reports variable definitions. The variables are reported in alphabetical order.

Variable	Definition
ATM ratio	The log of one plus the ratio of the aggregate AUM of institutions in a state to the total market capitalization of public firms in the same state.
CAPEX	Capital expenditures scaled by lagged assets.
CAPEX-to-sales	Capital expenditures scaled by sales.
CAPXRND	The sum of capital expenditures and research and development expenditures, all sc by lagged assets.
Cash flow	Net income before extraordinary items, depreciation and amortization expenses, all scaled by lagged assets.
Debt issuance	The change in assets, less the change in book equity, less the change in deferred taxes, scaled by lagged assets.
Equity issuance	The change in book equity and the change in deferred taxes, less the change in retained earnings, scaled by lagged assets.
Financial dependence	A firm is considered financially dependent if it meets three criteria: no dividend payout, no S&P crediting rating or S&P rating below BBB-, and sales not in the top quintile in the previous year among sample firms. A firm is classified as being non-dependent if any one of the three criteria is not met. Sales are defined as the ratio of net sales to lagged assets.
HKS ratio	The total book value of all firms located in state i to the aggregate income of all households located in state i.
INVESTMENT	The sum of total asset growth and R&D spending, all scaled by lagged total assets.
KZ index	The modified version of KZ index that omits Q following Baker et al. (2003).
Leverage	The sum of long-term debt and debt in current liabilities, all scaled by lagged asse
Log(Assets)	The log of total assets.
LP1	The log of the ratio of the aggregate weight of local firms in the portfolio of local institutions (local portfolio weight) to the aggregate weight of local firms in the m portfolio (local market weight).
LP2	The log of the ratio of the aggregate weight of local firms in the portfolio of local institutions (local portfolio weight) to the aggregate weight of local firms in the portfolio of all institutions (total portfolio weight).
New issuance	The ratio of the newly issued shares held by local institutions to the total number of newly issued shares.
Payout	The ratio of total distributions (dividends plus purchase of common and preferred stock minus preferred stock redemption value) to operating income.
R&D	Research and development expenditures scaled by lagged assets.
R&D-to-sales	Research and development expenditures scaled by sales.
Ret	Annualized stock returns.
ROE	Net income divided by lagged book equity.
Q	Market value of equity plus total assets, less book value of equity, all scaled by total assets.
StateQ	Value-weighted Q of firms within each state.
PIPCgrowth	The state-level personal income per capita growth rate.
Popgrowth	The state-level population growth rate.

ntary material

Baik, B., Kang, J.-K., Kim, J.M., 2010. Local institut
tion asymmetries, and equity returns. J. Finan

- A dot-com shock to local institutions lowers valuations of nearby non-tech firms.

7 Conclusion

7.1 Core conclusion

Institutional investor geography matters for corporate finance. Firms located near a large institutional investor base relative to local public equity supply have higher valuations and better equity financing access.

7.2 Economic intuition

Proximity lowers information acquisition and monitoring costs. When local institutional capital is abundant, firms have a deeper nearby base of informed equity providers.

7.3 Financing channel versus discount-rate channel

The evidence favors a financing channel. High-ATM firms issue more equity, invest more, and rely less on cash flow; ATM does not forecast stock returns.

7.4 Identification lesson

The causal interpretation is supported by the combination of mechanism tests, relocation tests, return tests, and the dot-com shock to institutional investors.

7.5 Broader implication

Investor-firm geographic dislocation can create real financial frictions across regions, even in a nationally integrated capital market.